Example 7.16

Find the angle of intersection of the curve $y = \sin x$ with the positive x-axis.

Solution

When the curve $y = \sin x$ intersects the positive x-axis, $y = 0$ which gives, $x = n\pi$, $n = 1, 2, 3, \ldots$.

Now, $\frac{dy}{dx} = \cos x$. The slope at $x = n\pi$ are $\cos(n\pi) = (-1)^n$. Hence, the required angle of intersection is

$$\tan\theta = \left|\frac{(-1)^n - 0}{1 + (-1)^n (0)}\right| = 1 \qquad \forall \text{ n}$$

Example 7.17

If the curves $ax^2 + by^2 = 1$ and $cx^2 + dy^2 = 1$ intersect each other orthogonally then,

$$\text{show that } \frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}.$$

Solution

Let the two curves intersect at a point (x_0, y_0). This leads to $(a-c)x_0^2 + (b-d)y_0^2 = 0$.

Let us now find the slope of the curves at the point of intersection (x_0, y_0). The slopes of the curves are as follows :

$$\text{For the curve } ax^2 + by^2 = 1, \quad \frac{dy}{dx} = -\frac{ax}{by} .$$

$$\text{For the curve } cx^2 + dy^2 = 1, \quad \frac{dy}{dx} = -\frac{cx}{dy} .$$

Now, if two curves cut orthogonally, then the product of their slopes, at the point of intersection (x_0, y_0), is -1. Hence, if the above two curves cut orthogonally at (x_0, y_0) then

$$\left(-\frac{ax_0}{by_0}\right) \times \left(-\frac{cx_0}{dy_0}\right) = -1.$$

$$\text{That is, } acx_0^2 + bdy_0^2 = 0,$$

$$\text{together with } (a-c)x_0^2 + (b-d)y_0^2 = 0$$

$$\text{gives, } \frac{a-c}{ac} = \frac{b-d}{bd}.$$

$$\text{That is, } \frac{1}{c} - \frac{1}{a} = \frac{1}{d} - \frac{1}{b}.$$

$$\text{Hence, } \frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}.$$

Remark

In the above example, the converse is also true. That is assuming the condition $\frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}$ one can easily establish that the curves cut orthogonally.

Chapter 7 Differential Calculus Original-new.indd 13 26-02-2020 17:22:22